PRODUCT REQUIREMENTS DOCUMENT

Increasing the average time spent on Spotify

Spotify India: Revenue & Free Tier User Experience

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Status: BrainStorming

Resources:

- Milestone 1
- Milestone 2
- Survey responses

Problem Definition:

What is the problem?

Free Spotify users frequently abandon their listening sessions due to **ad interruptions**, which negatively impacts their experience and leads to reduced session lengths and lower overall engagement with the platform.

Who are facing the problem?

This issue predominantly affects budget-conscious **free-tier** Spotify users in India, primarily aged **18-27.**

What is the value we can generate for Business by solving this issue?

Addressing this issue can boost **Time spent listening (TSL),** a critical engagement metric while improving retention, satisfaction, and Premium conversion rates. We can also preserves ad revenue by introducing alternative ad formats that balance user experience with advertiser objectives.

What value will the target user segment gain from solving this issue?

Users will control their ad experience by determining when and how ads play, creating uninterrupted, higher-quality listening sessions while staying on the Free Plan.

How do we know this is the real problem?

Ad interruptions negatively impact user experience, resulting in reduced session lengths and increased app abandonment. This engagement decline directly affects key business metrics: lower ad revenue, decreased Premium conversions, and diminished retention rates in India's competitive streaming environment.

Why Should we solve this problem now?

India's streaming landscape is growing more competitive as JioSaavn integrates with telecom services, and Gaana and YouTube Music offer frictionless listening alternatives. Addressing free-tier engagement is vital given mounting user expectations and widespread ad interruption fatigue.

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Key Goals & Success Metrics:

Key Goals	Metrics to track	Type	Why it Matters
Increasing Total Listening Time	• Avg Listening time per Session • Avg #Sessions per Day	Functional Metrics	Directly reflects the business outcome & user engagement
Increasing Revenue	• Adoption Rate of the new plan • New Plan Retention rate • New Plan to Premium conversion rate	Functional Metric	Reflects the adoption of new plan & also ensures that it's not impacting other parts of the business.
Ensuring Ad Monitization	• Ad Completion Rate • Session Drop-off Rate at ads	Functional Metric	ACR directly reflects the Ad Monetization & session drop-off shows user frustration and impacts ad revenue
Improving User Satisfaction	• NPS/CSAT for ad experience	Non Functional Metric	Shows User Sentiments towards the Product

Why are these metrics important?

These metrics measure our primary goal of driving higher time spent listening through improved engagement and retention, while maintaining user satisfaction and experience alongside stable ad revenue and business performance.

Non Goals:

The Areas where we are not planning to do any changes:

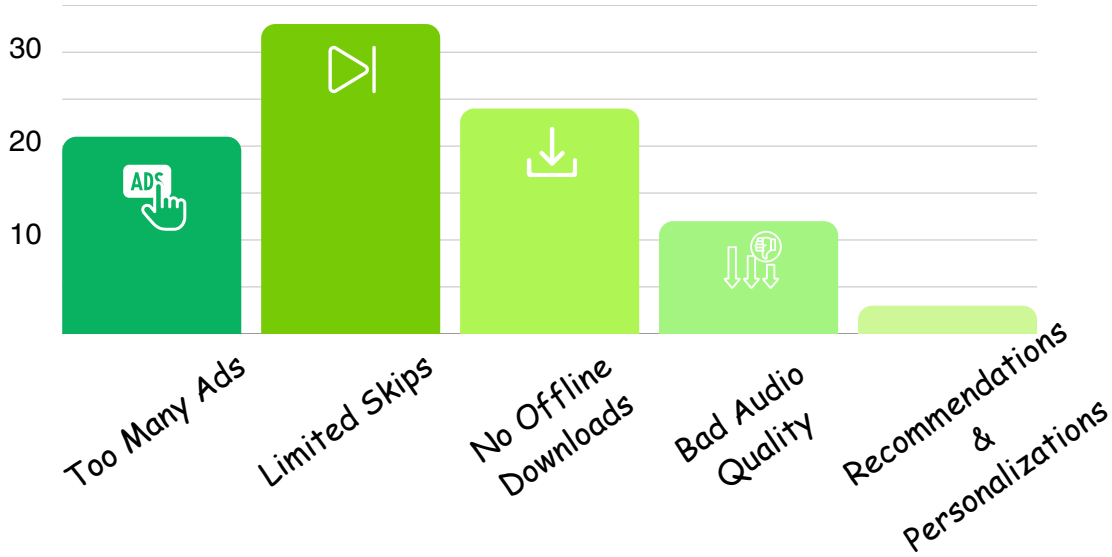
- Removing Ads Completely.
- No Changes to current Premium Plans or features.
- No significant impact to the current Ads Revenue.
- Providing all the premium plan features to the new plan users.

Validation of the Problem:

Insights from User Research:

- In a user survey, in which 36 free-tier users participated, it was revealed that 83% of them are frustrated by the **lack of song control** and **excessive, random ad interruptions**.
- Many respondents reported abandoning sessions or discontinuing app use due to random ads

Key Pain Points for Free Users



- The survey data validates that the core of user frustration with the free tier stems directly from a lack of control over their listening experience and the frequent, disruptive nature of ad interruptions.

Key Insights from Competitive Landscape:

- India's income levels limit willingness to pay premium subscription fees, users prefer smaller, budget-friendly amounts instead. Competitors have adapted accordingly:
  - **JioSaavn** dominates with 100M+ MAUs through telecom bundled plans at ₹10 to ₹20 monthly and generous free tiers.
  - **YouTube Music** and **Gaana** use "watch ad for uninterrupted listening" models plus affordable micro plans, balancing revenue with user satisfaction.

Understanding the Target Audience:

- Spotify's Free-tier users in India, aged 18-27, comprise a highly engaged, mobile first, cost-conscious audience.
  - MAUs of Spotify India (2025)~ 83 Million
  - Free tier MAUs of Spotify India~ 80 Million

Assuming 20%of the population lies in the 18-25 age segment:

- Size of Target user Segment~ 20% of 80 Million = **16 Million**

User Personas:

Ayushi, 19 | Student | Delhi

- Daily listener during study sessions, commutes, workouts, and chores; drops off from sessions when random ads interrupt without notice.
- Disruptive random ads interrupt her focus. Limited song control frustrates her listening experience.

Ayush, 26 | SDE | Bangalore

- Listens while working, commuting, and exercising
- Sudden ads break deep focus and disrupt flow during work sessions
- Prefers watching ads upfront to earn ad-free time and see when the next ad comes, allowing strategic planning.

User Journey Map:

Stage:	Session Initiation	Discovery	Early Listening	First Ad Break	Post Ad Session
Steps:	Opens Spotify Ad pops out on home screen	Browses suggestion & playlists Chooses a recommended playlist	Starts Listening on smart shuffle Want to skip current song but not able to do it Likes this song, want to listen this song again but can't play it again	Music stops randomly and audio ads begins	Continues listening again Want to play a specific song, searched & found the song but can't play it.
Thoughts:	"I prefer listening music while coding or during work time" "I listen to music every time while commuting"	"I usually go to explore What's new section to explore some fresh songs" "I usually go to You might also like section"	"This recommended playlist is pretty good" "I don't like it, Why can't I skip it?" "This is my favorite, I wish I could listen this in loop"	"I was enjoying my playlist, now my flow got interrupted. I just hate these random Ads!"	"Finally these Ads are over." "This song just came to my mind, I searched if it's available it was but i can't play it immediately"

Emotions:

- **Neutral** — Wants to listen music while working
- **Excited** — Wants to explore some new music
- **Annoyed** — Couldn't explore new song that matches his taste at all
- **Happy** — Liked a new song from the playlist
- **Disappointed** — Wants to replay song but can't
- **Surprised** — Smart Shuffle is working well, and user is enjoying music while working
- **Frustrated** — Audio Ad started, session got interrupted randomly
- **Relaxed** — Finally ad is over and session continued

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User's Pain Points:

- Desire for control over music and relief from random ads without paying a substantial amount.
- Wants to know when ads will play and have control over ad timing.
- Annoyed by unpredictable ad interruptions and lack of playback control.

Solution Overview:

Solution 1: Spotify Mobile (Ad enabled Premium Plan)

Similar to JioHotstar Mobile Plan

Key Idea:

Low-price subscription featuring reduced ad frequency, ad transparency, and additional features. (Only for Mobile)

Working:

Launch a ₹29/month plan offering limited ads (2 per hour), complete playback control, and offline mode access.

User Benefits:

Affordable plan for getting access to

- Unrestricted song selection control
- Reduced interruptions with ad transparency
- Offline listening for network-free access

Target Segment:

- Price sensitive free tier users (~4M approx.)

Key Features of Spotify Mobile Plan:

- Offline Mode access
- Play song in any order
- Listen with friends in real time
- Organise Listening Queue
- Listening Insights (Your Sound Capsule)
- Only 2 ads per hour with ad transparency

Validation of Mobile Tier Opportunity

Survey data from 35 free-tier Spotify users (n=53 total responses) reveals that 83.3% prefer a Mobile subscription model over the current ad-supported free tier, validating demand for an affordable option that offers better control and fewer interruptions.

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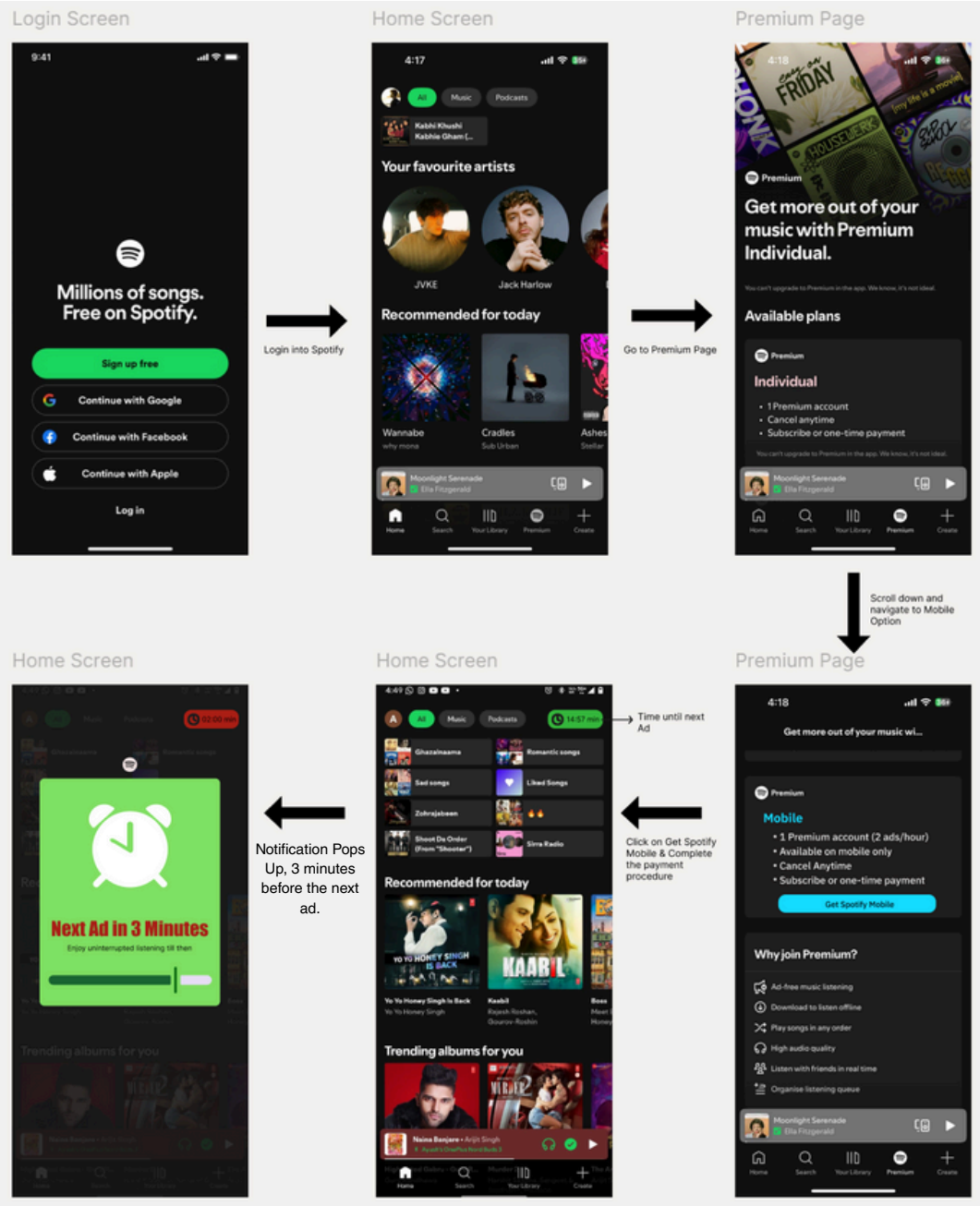
User Flow & Wireframes:



Login to Spotify Mobile

↓

Go to Premium Page

↓

Scroll down to Spotify Mobile Option

↓

Click on Get Spotify Mobile

↓

Complete the Payment Procedure

↓

Enjoy Listening sessions with Spotify Mobile

↓

You can see time countdown for next ad at the top right of home screen.

UX Behaviour Specification:

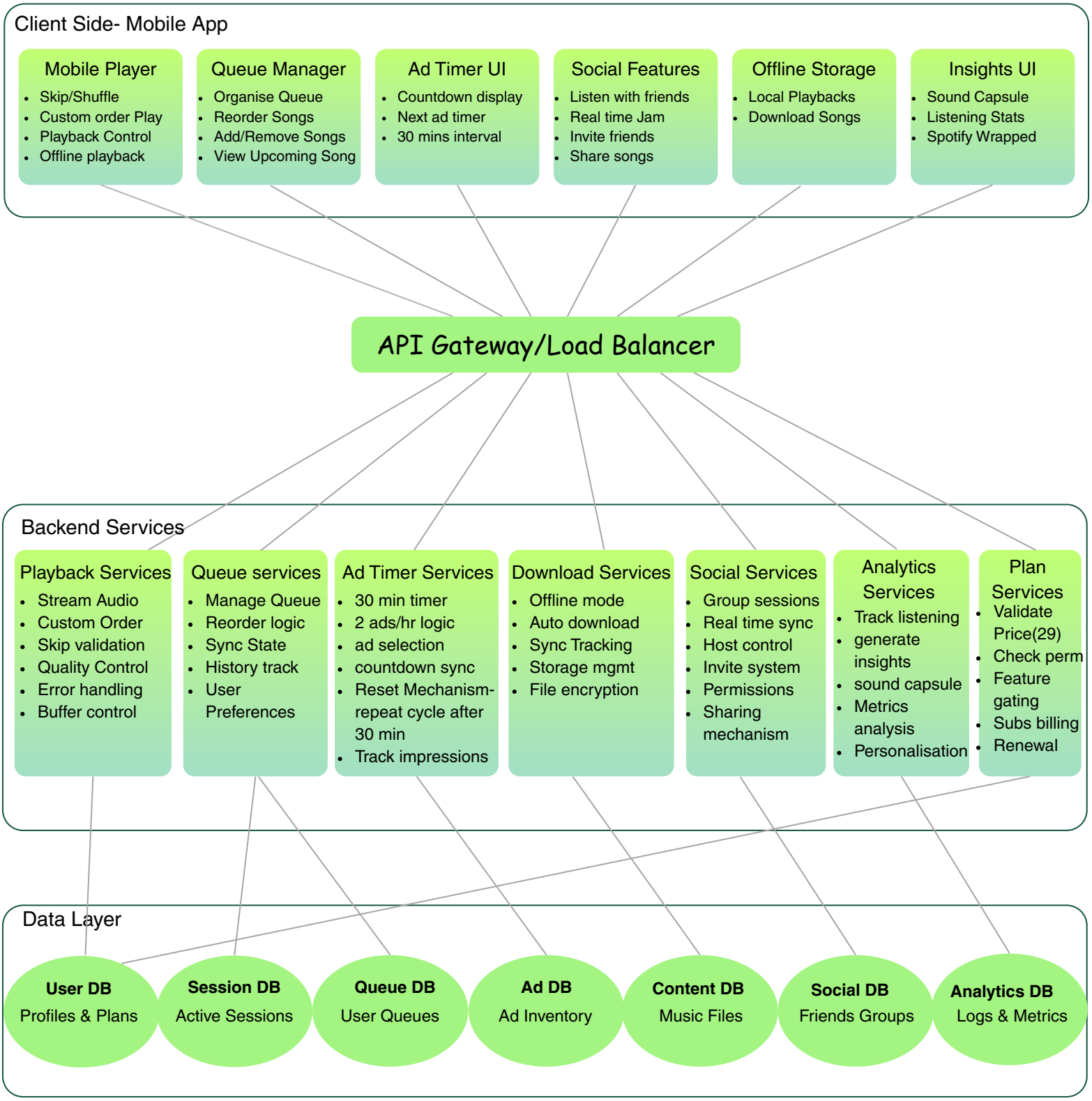
Ad Timer:

State	Trigger	Behavior
Active Countdown	Playback in Progress	Displays “XX:YY” (mm:ss format), updating every second
Final Countdown (30 sec)	30 Sec before ad	Timer text changes to Red And a popup comes “Next Ad in 30 sec”
Post-Ad Cooldown	Ad Finishes	Popup Shows Enjoy Ad Free listening for next 30 minutes. Timer restarts
Paused Playback	Listening Session paused	Timer paused, restarts upon session restarts
Offline Mode	user is offline	Timer Hidden, ads suspended till network reconnection.

Backend Architecture

- **New Plan Type:** Introduce SPOTIFY_MOBILE subscription tier restricted to mobile devices only.
- **Events to track Ads:**
  - Ad_Break_Started: Fired when scheduled ad break begins
  - Ad_Break_Completed: when ad break finishes and music resumes
  - Ad_timer_paused: When user pauses listening session
  - Ad_frustration_exit: When user discontinues session in between of an ad
  - Ad_break_error: When ad playback fails
- **Events to track Conversion:**
  - free_to_mobile: To track Mobile Plan Subscription, will trigger after succesfull purchase of plan.
  - premium_to_mobile_downgrade: To Track Premium users who shifted to Mobile Plan.
- **Schema Changes:**
  - Add new plan to subscription_plans table with fields for device_restriction (mobile only), ad_frequency (2 per hour)
  - Modify user_sessions table to include device_type, is_active_session, and device_id fields.
  - Create ad_quota_tracker table logging user_id, ads_served_count, hour_window_start, and next_ad_timestamp
  - Add feature_entitlements table mapping plan_type to allowed features with device restrictions.
- **UI Implementations:**
  - Timer displays "[MM]" with color coding (green/yellow/red based on time remaining).
  - Display device switch alert: "Already logged on [Device]. Switch to this device?" with confirmation required.
- **Feature Gating:**
  - Enable premium features (manual play, unlimited skips, offline downloads) on mobile devices only.

System Diagram:



Data Instrumentation Plan

Events	Key Fields	Purpose
Free_to_mobile	user_id, device_type, previous_plan = free, new_plan = mobile, Conversion_channel	Captures conversion from Free Plan → Spotify Mobile Plan. Used for activation and revenue attribution.
Premium_to_mobile	user_id, device_type, previous_plan = premium, new_plan = mobile	Tracks downgrades from Premium to Mobile Plan. Used for churn detection and downgrade analytics.
Ad_break_started	user_id, timestamp, ad_break_id, device_type	Log the start of an ad break (sequence of ads)
Ad_break_ended	user_id, timestamp, ad_break_id, device_type completion_status	Captures successful completion of the entire ad break (all ads played). Used for funnel analysis and monetization tracking.
Ad_timer_paused	user_id, timestamp, ad_break_id, pause_reason	Log when the user pauses the Listening session or switch to offline mode
Ad_frustration_exit	user_id, timestamp, ad-id(if mid ad), ad_break_id	Track when a user exits, closes app, or navigates away during or right before an ad break. Helps track CSAT and optimize UX.
Ad_break_error	user_id, timestamp, ad_break_id, error_type, network_type, device_type	Logs any playback, buffering, or quota-related errors during an ad break. Critical for debugging.

Edge cases handling:

Edge Case	Expected Handlings
Ad fails to load (no-fill)	Skip ad → resume music → do NOT count toward hourly quota → log error
Ad asset loads but cannot buffer due to weak network	Retry once → if still failing → skip ad → resume content
User exits app during ad	Log frustration exit → do not count ad → on return evaluate eligibility again
App crashes during ad	On restart treat ad as incomplete → do NOT count toward quota
Ad error just after play starts	Retry playback from start once → if error again → skip
Timezone change	Always use server time → prevent double-serving ads
User toggles airplane mode repeatedly	Limit retry attempts → skip after 2 retries → avoid loop
User tries switching device or reinstalling app	Ad-Timer tied to Account not device

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Launch Readiness & Key Milestones:

Designing Phase
2-3 Weeks

Finalize:

- Wireframes & UI for ad timer
- User Journey to check and maximize user satisfaction

Development Phase
1 Month

Build:

- Subscription & Device Management System
- Ad Frequency Controller with Timer UI
- Feature Gating & Payment Processing

QA & Internal Testing
1 Month

Test:

- Login to multiple account.
- Correct working of timer
- login to other devices except mobile
- Usability/any hinderance of User Flow

Dogfooding/Beta launch
2 Weeks

- Internal Rollout of new plan in some tier 1 cities.
- Collect feedback on the UI/ User journey and Iterate

Marketing Prepartation
3 Weeks

- Designing in-app banners & Push Notifications
- Aligning with Marketing team for proper product positioning.

Launch Checklist:

Internal Stakeholder:

- Product, Engineering, QA, Analytics
- Legal & Compilence
- Marketing, Customer Support

Product & Technical Readiness

- Build device authentication
- single-device enforcement
- UI Implementation: Timer
- Analytics Tracking Setup
- Payment Processing Integration
- Onboarding flow and FAQ documentation

Marketing Alignment

- Targeted ads to Free users
- Rollout campaign scheduled

Customer Support

- FAQs & customer help doc
- Training of Support team for new feature rollout

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Launch Checklist:

QA Checklist:

- Plan Activation Testing
- Payment Methods & Billing
- Ad Frequency Testing
- Playback Controls Testing
- Queue Management Testing
- Ad Playback & Transparency Testing
- Performance Testing
- UI, User Flow & Responsiveness Testing
- Device Type Testing
- Social Features Testing
- Offline Mode Testing
- Download feature Testing
- Listening Insights- Data Collection & Capsule display Testing
- Security & Privacy Testing

Future Iterations:

V2: Rewarded Ads Model

- Users choose when to watch ads
- Earn uninterrupted listening time
- Same ad frequency (2 per hour) = consistent revenue
- Voluntary, predictable engagement

Open Questions & Decision Taken:

Ideal Mobile plan price to drive adoption, remain competitive, and ensure profitability?

➢ Pricing is set at ₹29/month to attract price-sensitive younger users while maintaining profitability for Spotify.

How much control will users have over ad frequency and timing?

➢ Initial release limits ad control to reduced frequency and predictable placement. No ad skipping or type selection, in the first iteration focus is on fewer interruptions and transparency.

Will Premium subscribers downgrade to Mobile Plan?

➢ Minimal Premium churn expected. Ad-free listening and high-quality audio are habit-forming, users won't willingly return to ads. Premium-to-Mobile conversion will be monitored as a safeguard metric.

Risk & Mitigations:

⚠ **Risk: Users may expect fully ad-free experience despite Mobile plan retaining some ads.**

Upfront communication emphasizing "significantly fewer ads" and in-app messaging to clarify tier benefits versus limitations.